

COURSE TITLE : GENERAL CHEMISTRY LAB
COURSE CODE : 3078
COURSE CATEGORY : B
PERIODS/WEEK : 6
PERIODS/SEMESTER : 90
CREDIT : 3

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Permanganometry	22
2	Dichometry, Iodometry	22
3	Argentometry, Gravimetric Analysis	22
4	Physical Chemistry Exercises Organic Preparations	24
TOTAL		90

COURSE CONTENT

MODULE– I

1.1.0 Permanganometry

- 1.1.1 Standardization of KMnO_4 using std. Mohr's salt solution
- 1.1.2 Estimation of Ferrous Iron using std. Oxalic acid Estimation of Oxalic acid using Mohr's salt crystals Estimation of Ca using Oxalic acid crystals

MODULE – II

2.1.0 Dichometry

- 2.1.1 Preparation of standard solution of Potassium dichromate
- 2.1.3 Estimation of Ferrous Iron using internal indicator
- 2.1.4 Estimation of Ferrous Iron using external indicator

2.2.0 Iodometry

- 2.2.1 Standardisation of sodium thiosulphate solution
- 2.2.2 Estimation of Copper
- 2.2.3 Estimation of Potassium dichromate
- 2.2.4 Estimation of available Chlorine in bleaching powder

MODULE– III

3.1.0 Argentometry

- 3.1.1 Estimation of chloride ion (Mohr's method)
- 3.1.2 Estimation of chloride ion in acid medium (Mohr's method)
- 3.1.3 Estimation of Potassium bromide

Gravimetric Analysis

Estimation of Ba^{2+} as BaSO_4

Estimation of SO_4^{2-} as BaSO_4

Estimation of Fe^{3+} as $\text{Fe}(\text{OH})_3$

MODULE – IV

4.1.0 Physical Chemistry Experiments

- 4.1.1 Determination of melting point and boiling point
- 4.1.2 Determination of the critical solution temperature of phenol and water
- 4.1.3 Determination of elevation in boiling point
- 4.1.4 Determination of the depression in freezing point

Organic Preparations

Dinitrobenzene starting from nitrobenzene – (Nitration)

Acetanilide starting from aniline (Acetylation)

Benzanilide from aniline (Benzoylation)

Benzoic acid from methyl benzoate (Hydrolysis)

Benzoic acid from benzyl chloride (side chain oxidation)

REFERENCE BOOKS

1. Practical Chemistry for B. Sc Main – A. O. Thomas and Mani.